Team Tokenizers — One Encounter, Two Readers: Verified Clinical Notes for Doctors and Faithful Summaries for Patients

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Abstract

A clinical intake system produces one record that two people read: the doctor signs a SOAP note, and the patient receives a plain-language summary. Current LLM systems collect the evidence first and verify the record afterwards. We argue that this separation causes two failures. **Unrecoverable gaps:** verification finds a missing or unsupported fact only after the patient, the one source who could supply it, has left the conversation, so the fact can only be deleted or guessed. **A fact–inference category error:** faithfulness filters test every sentence for entailment, but diagnoses and plans are inferences that no dialogue turn entails, so correct reasoning is deleted. We make both failures measurable with known answers (withheld facts, planted errors, valid assessments) and a *checkpoint-branching* protocol in which every compared strategy continues from the identical intake dialogue. We then test two minimal fixes, *verify-then-ask* and *section-aware verification*; all other components are held fixed and evaluated only as supporting parts.

1 Problem and Motivation

Problem statement. LLM agents can take a patient history and write a clinical note, but the note may contain facts the patient never stated and omit facts the patient would have reported if asked. Interactive intake systems such as Aegle (Chen et al., 2026) reduce omissions by asking better questions, but do not check individual statements in the note. Post-hoc filters such as uMedSum (Nagar et al., 2025) check atomic facts, but only after the source is fixed, and with one entailment test for every sentence. Chaining the two leaves a structural gap: when the filter finds that a mandatory fact, such as an allergy, is missing or unsupported, it can only delete the statement or paste in similar text, because the patient can no longer be asked. The filter also cannot tell a reported fact ("chest pain for two hours") from a clinical inference ("likely stable angina"); the inference is never entailed by the dialogue, so a correct diagnosis is removed. Both errors reach both readers: the doctor signs an incomplete or over-pruned note, and the patient's summary inherits it.

Why it matters. The first-visit note anchors later decisions, so a dropped allergy or a deleted correct assessment propagates through care. The doctor's review effort depends on how much of the note must be re-checked, and the patient's summary carries the clinic's authority even when it is wrong.

Domain and target users. Healthcare, restricted to adult outpatient first-visit intake and the SOAP Initial Progress Note. The *patient* answers plain-language questions and, after sign-off, receives a lay summary; red-flag symptoms trigger an immediate alert. The *doctor* reviews flagged items and signs a note in which every statement links to its evidence.

What is missing in existing approaches. Intake agents (Chen et al., 2026) are evaluated mainly with rubric scores, not statement-level checks. Atomic-fact and NLI-based checking (Min et al., 2023; Laban et al., 2022; Nagar et al., 2025) and self-refinement (Madaan et al., 2023) assume the source is complete and fixed, and treat every sentence as extractive. Clarifying-question research (Alianne-jadi et al., 2019) shows that asking resolves uncertainty, but it is not triggered by claim-level verification. No prior work (i) uses a verification result to ask the source while it is still available, or (ii) separates facts from inferences when deciding what counts as supported.

2 Research Questions and Hypotheses

RQ1 (unrecoverable gaps). At a matched budget of patient turns, does asking a targeted question when verification finds a gap recover more missing

facts, without lowering faithfulness, than deleting, guessing, or asking more untargeted questions?

RQ2 (category error). With the same verifier model, does checking Assessment/Plan (A/P) statements by their evidence chain instead of by entailment keep more correct reasoning while still rejecting unsupported diagnoses and plans?

- **H1a:** Verify-then-ask recovers more withheld facts than delete-only and than post-hoc filtering.
- **H1b:** It recovers more than the same number of untargeted extra turns, so *targeting* drives the gain.
- **H1c:** Its grounding rate is at least that of post-hoc filtering, while guessing lowers grounding.
- **H2:** Uniform entailment rejects many valid A/P statements even with our verifier; the evidence-chain rule rejects fewer while keeping recall on planted unsupported A/P statements within 5 points.
- **H3 (secondary):** Patient summaries built from the verified record contain fewer unsupported statements at similar reading level.

3 Capabilities: Core vs. Supporting

Two capabilities answer the research questions. The others are implementation choices; each has exactly one check, and we claim nothing beyond it (Table 1). Multi-agent intake does not test either research question—it only produces the dialogue that verification acts on—so it is held fixed for every comparison and varied once only, to test whether the RQ1 gain depends on intake quality.

4 Proposed Approach

Figure 1 shows one encounter. Notation: dialogue turns $D = (d_1, \ldots, d_T)$; SOAP features $F = \{f_i\}$, each linked to its source; atomic claims $C = \{c_j\}$ from the draft note.

Fixed intake. We reuse Aegle's design: an orchestrator activates up to three specialists, who propose questions in parallel without seeing each other, and an aggregator writes the state before asking. Intake stops when mandatory fields are filled, or three turns before the cap. This point is the *checkpoint*.

Verify-then-ask (RQ1). The draft is split into atomic claims, and each claim is checked against evidence from its own section: Subjective claims against the top-k retrieved patient turns, Objective claims against the encounter's measurement record (vitals and exam values, which a patient cannot report). A verifier labels each claim *supported*, *contradicted* or *unsupported*, and fixed controller rules act: supported → keep; contradicted with the correct value in a cited turn → correct from that turn and re-verify; an unsupported or missing Subjective fact about a symptom, medication, allergy, duration or mandatory field → ask one targeted question; an Objective gap → flag for the doctor; anything else → delete. The answer is re-verified, and the claim is kept with the new turn as evidence, deleted if denied, or marked "patient unsure". At most three questions are asked. Rules, not the LLM, choose the action, so every action is reproducible and logged with its reason.

Section-aware verification (RQ2). After the freeze, specialists write the Assessment and Plan, and each statement must cite the features it rests on. Because A/P statements are inferences, the verifier does not test entailment; it judges whether the cited features, together with a retrieved reference passage when one is found, justify the statement (*chain_ok* / *chain_broken*). A broken chain is flagged for the doctor rather than deleted, since removing a diagnosis is a clinical decision.

Two readers. The doctor receives the note with each statement linked to its evidence, plus the flags. After sign-off (simulated in experiments), a patient summary is generated from the signed note, excluding any item still flagged; each summary sentence is verified against the signed note before release.

Supporting components. *Verifier:* one model for every check, Qwen2.5-3B-Instruct with QLoRA (Dettmers et al., 2023) (rank 16), trained on ∼6k claim–evidence pairs from MTS-Dialog (Ben Abacha et al., 2023) and ACI-Bench train with six perturbation types; S/O and A/P use different prompts and label sets. *Retrieval:* BM25 and dense embeddings fused by reciprocal rank (Cormack et al., 2009); $k \in \{1, 3, 5\}$ is tuned on dev for evidence recall. *Reference passages:* MedlinePlus topic pages (public domain) for the test complaints, chunked by section; these are consumer-level references, not clinical practice guidelines. *Tools:* guards block writes after the freeze, diagnosis or treatment wording to the patient before sign-off, finalising with an unresolved red flag, and release before sign-off. All agents use Qwen2.5-7B-Instruct.

Capability	Role	Baseline / ablation that isolates it	Metric
Agentic control	**Core, RQ1**	Decides keep, correct, ask, or mark unsure per claim. Compared with delete-only, guess-only, extra untargeted turns and post-hoc filtering, all from the same checkpoint; question-cap sweep	Withheld-fact recovery, critical-fact recall, grounding rate, patient turns
Reasoning	**Core, RQ2**	Claim decomposition and section-aware checking. Compared with uniform entailment using DeBERTa and using our own verifier; intermediate verdicts scored	Valid-claim retention, unsupported-A/P rejection precision/recall, per-step accuracy
Tool calling	Supporting	Typed tools and safety guards; blocked attempts are logged, so guards-off is measured without an extra run	Argument validity, unsafe-action rate, missed red flags
RAG	Supporting	Top-k turn retrieval vs. whole dialogue; A/P check with vs. without reference passages	Evidence recall@k, citation precision
Fine-tuning	Supporting	QLoRA 3B verifier vs. DeBERTa NLI vs. prompted 7B on planted-error claims	Macro-F1 per error type, seconds per claim
Multi-agent intake	Held fixed	Single-agent intake, robustness of RQ1 only	RQ1 metrics; no separate claim

Table 1: Core capabilities answer the research questions; each supporting capability has one check; multi-agent intake is a fixed substrate.

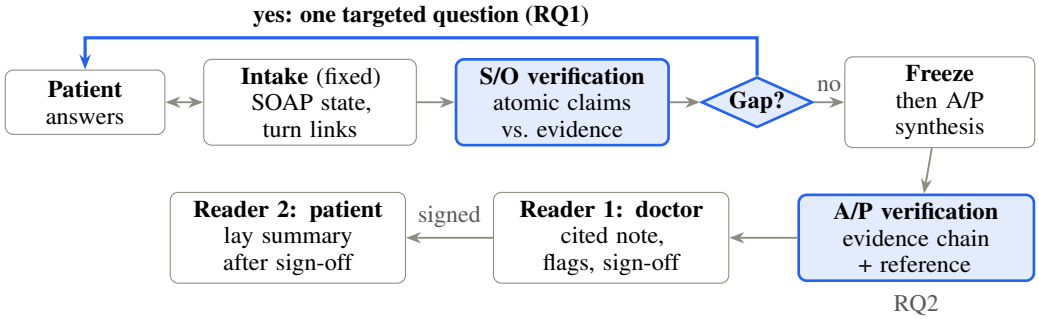


Figure 1: One encounter. Shaded parts are tested. A gap found before the freeze becomes a question to the patient (RQ1); after the freeze, A/P statements are checked by their evidence chain (RQ2). Both readers' outputs come from the same verified record.

5 Evaluation Plan

Data and known answers. All data are public or synthetic; no real patient takes part. **E1 (40 encounters + 10 dev):** ACI-Bench (Yim et al., 2023) test1 (dev: valid). An LLM simulated patient answers only from a hidden case file of atomic facts, checked by two team members. In each case two mandatory Subjective facts are *withheld*: revealed only to a direct question. Objective facts form the measurement record; eight cases contain a planted red flag. **E2 (120 notes):** drafts from E1 reference notes, 90 with one planted error (fabricated symptom, negation flip, medication or dose, laterality or duration, unsupported diagnosis, plan without evidence; 15 each) and 30 unchanged; the reference A/P statements form the *valid* set.

Checkpoint branching and baselines. Intake runs once per case and seed and is saved at the checkpoint. Every RQ1 condition continues from that saved state with the same note writer and at most three more patient turns, so comparisons are paired on identical dialogues. Every RQ2 condition receives the same frozen A/P statements and is scored on its *reject* decision (flag or delete), so our rule cannot win simply by never deleting. Table 2 lists all conditions.

Metrics. *RQ1:* withheld-fact recovery (withheld facts present and supported in the final note); atomic and critical fact recall (allergies, medications, red-flag symptoms); S/O grounding rate; hallucination on gaps; patient turns. Diagnostics to explain outcomes: gap-detection precision/recall, gaps resolved per question, redundant-question rate. *RQ2:* valid-claim retention (valid A/P statements not rejected) and unsupported-A/P rejection precision/recall; per-step verdict accuracy. *Patient reader:* unsupported statements per summary, recall of next steps and warning signs, Flesch–Kincaid grade. *Supporting:* verifier macro-F1 per error type, evidence recall@k, citation precision,

ID	Condition	What it isolates	Tests
Baselines: existing approaches (same backbone, budgets, simulated patients)			
B1	Single-agent intake → note, no verification	Plain LLM, own intake	context
B2	Aegle-style intake → note, no verification (from the checkpoint)	Intake alone	RQ1
B3	B2 → uMedSum filter: NLI deletion, coverage-based insertion, published thresholds	**Main baseline**: verify after intake	RQ1, RQ2
B4	B2 → one round of self-refine, no tools	Unstructured self-critique	RQ1
RQ1 controls: each removes one ingredient of verify-then-ask, from the same checkpoint			
C1	In-loop verification; unsupported claims deleted, gaps left empty	Is *asking* needed?	H1a
C2	In-loop verification; gaps filled by the LLM without asking	Asking vs. guessing	H1c
C3	Same number of extra turns as ours, questions chosen by the intake policy	Targeted vs. untargeted	H1b
C4	Ours with question cap 1, 2, 3, 5	Completeness vs. turns	H1
RQ2 conditions: same frozen A/P statements			
U1	Uniform entailment with DeBERTa NLI (uMedSum thresholds)	Published approach	H2
U2	Uniform entailment with *our* verifier	The rule, with the model fixed	H2
U3	Evidence chain without reference passages	Contribution of retrieval	support
Supporting checks			
S1–S4	Verifier swap (fine-tuning); whole dialogue instead of top-k (RAG); single-agent intake (robustness); summary from unverified record (H3)	One check each	support

Table 2: All compared conditions. RQ1 conditions share the intake checkpoint; RQ2 conditions share the frozen A/P statements.

tool argument validity. *Safety:* unsafe-action rate (advice to the patient before sign-off, release before sign-off, write after freeze, finalising with an unresolved red flag) and missed red flags. Primary metrics come from known answers or rules; an LLM judge is used only for fact presence and claim support on E1 notes and is validated on 30 double-annotated notes (Cohen's $\kappa \geq 0.6$), otherwise those metrics become secondary.

Experiments and statistics. **Exp. 1 (the problem):** on B2 and B3, the share of withheld facts missing at the checkpoint and still missing after post-hoc repair, and the valid-claim retention of U1. **Exp. 2 (RQ1):** ours vs. B3 and C1–C4 on E1, 2 seeds. **Exp. 3 (RQ2):** U1–U3 vs. ours on E2 and E1 A/P statements. **Exp. 4:** supporting and safety checks. Seeds are averaged per case; 95% intervals come from a paired bootstrap over cases (Berg-Kirkpatrick et al., 2012); exact McNemar tests are used for per-fact recovery, with Holm correction across primary metrics. For each comparison we fix in advance the diagnostic that explains the outcome either way: for example, if ours does not beat C3, gap-detection recall and question yield show whether the verifier missed the gaps or the questions failed to fill them. A failure analysis assigns 30 errors to the first stage that went wrong.

6 Expected Novelty

1. **A named, measured failure of the intake-then-verify design:** verification that comes too late to fix a gap, and entailment as the wrong test for clinical inferences, shown with known answers rather than rubric scores.
2. **A reusable evaluation protocol:** checkpoint branching with withheld facts compares any verification strategy on identical dialogues.
3. **Evidence that minimal fixes suffice:** two small, explainable changes, each tested against a control that removes only that change.

7 Compute Plan

GTX 1080 Ti GPUs (compute capability 6.1) are not supported by vLLM's standard builds, so the 7B backbone and the simulated patient run through OpenRouter (Qwen2.5-7B-Instruct, listed at US$0.10/$0.20 per million input/output tokens); ACI-Bench is public, so this is permitted. The cluster runs verifier training and inference, DeBERTa NLI and retrieval. Kaggle and Colab are backups.

8 Expected Risks and Fallback Plan

Safety and limitations. No real patients take part. The patient-facing agent cannot state a diagnosis or treatment before sign-off, red flags trigger immediate escalation, and every action is logged. Sim-

Item	Estimate
Verifier QLoRA (3B, 4-bit, fp16), 2 runs	8 GPU-h
Verifier, NLI and retrieval in all runs	12 GPU-h
Development and debugging	8 GPU-h
GPU total (1080 Ti)	**28 GPU-h**
Backbone, ∼40M tokens (intake reused by all branches)	US$5
LLM judge and validation	US$15
API total, hard cap	**US$25–50**

Table 3: Compute budget.

Risk (likelihood)	Fallback
Verifier misses gaps (M)	Report gap-detection recall; rerun with oracle gaps to separate detection from asking
Withheld facts leak to general questions (M)	Audit dev dialogues; score only facts still missing at the checkpoint
Simulated patient invents facts (M)	Case-file-only prompt, temperature 0.1, audit; drop failing cases
Tuned verifier not better (M)	Use the best on dev; RQ1 and RQ2 are unaffected
Reference passages miss topics (H)	Evidence chain alone (U3) becomes the main condition
Limited annotation time (H)	30 double-labelled notes; primary claims rest on known answers

Table 4: Risks and fallbacks (H/M = high/medium).

ulated patients are more consistent than real ones, withheld facts approximate real omissions, data are English role-played encounters, and sign-off is simulated. A negative result on either question is still informative, and the fixed diagnostics are designed to explain it.

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